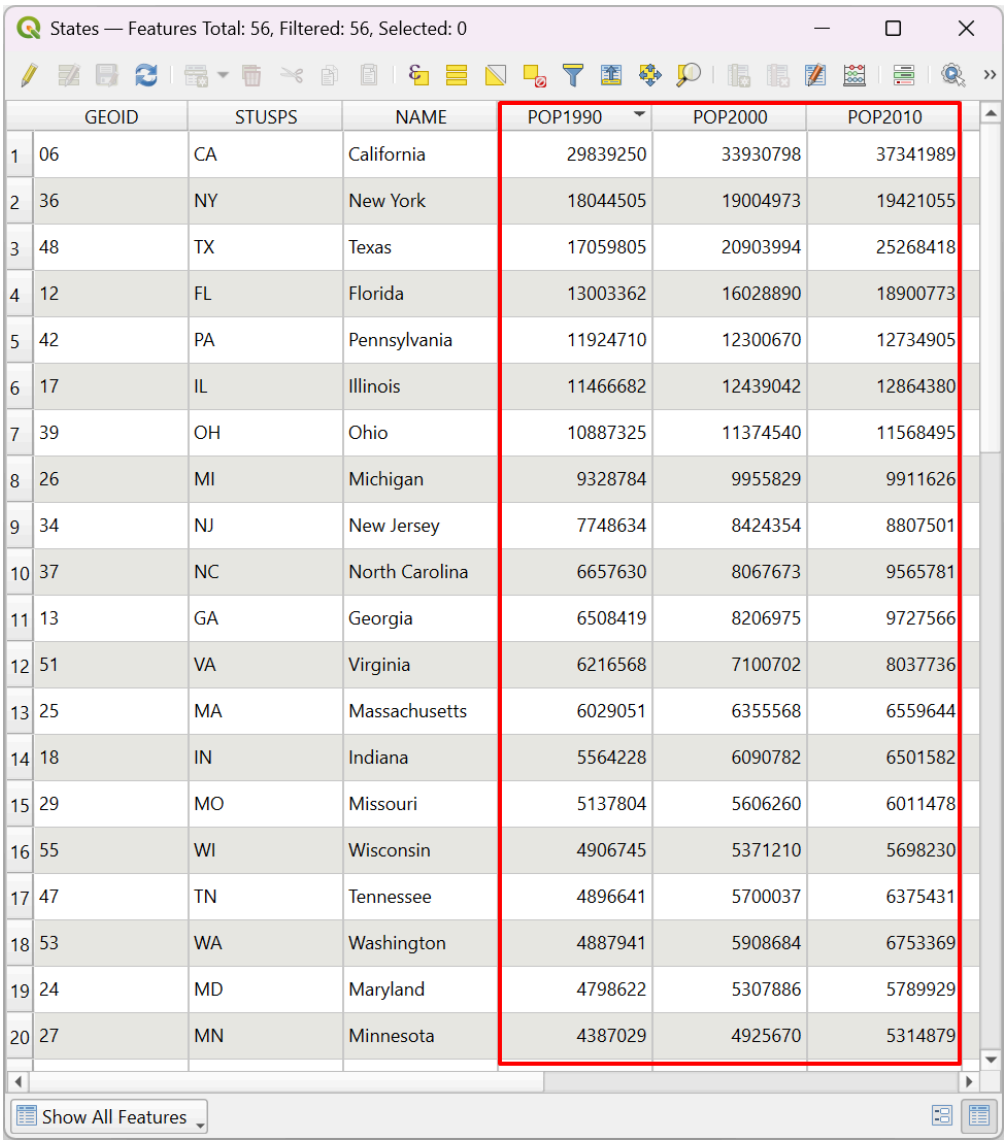
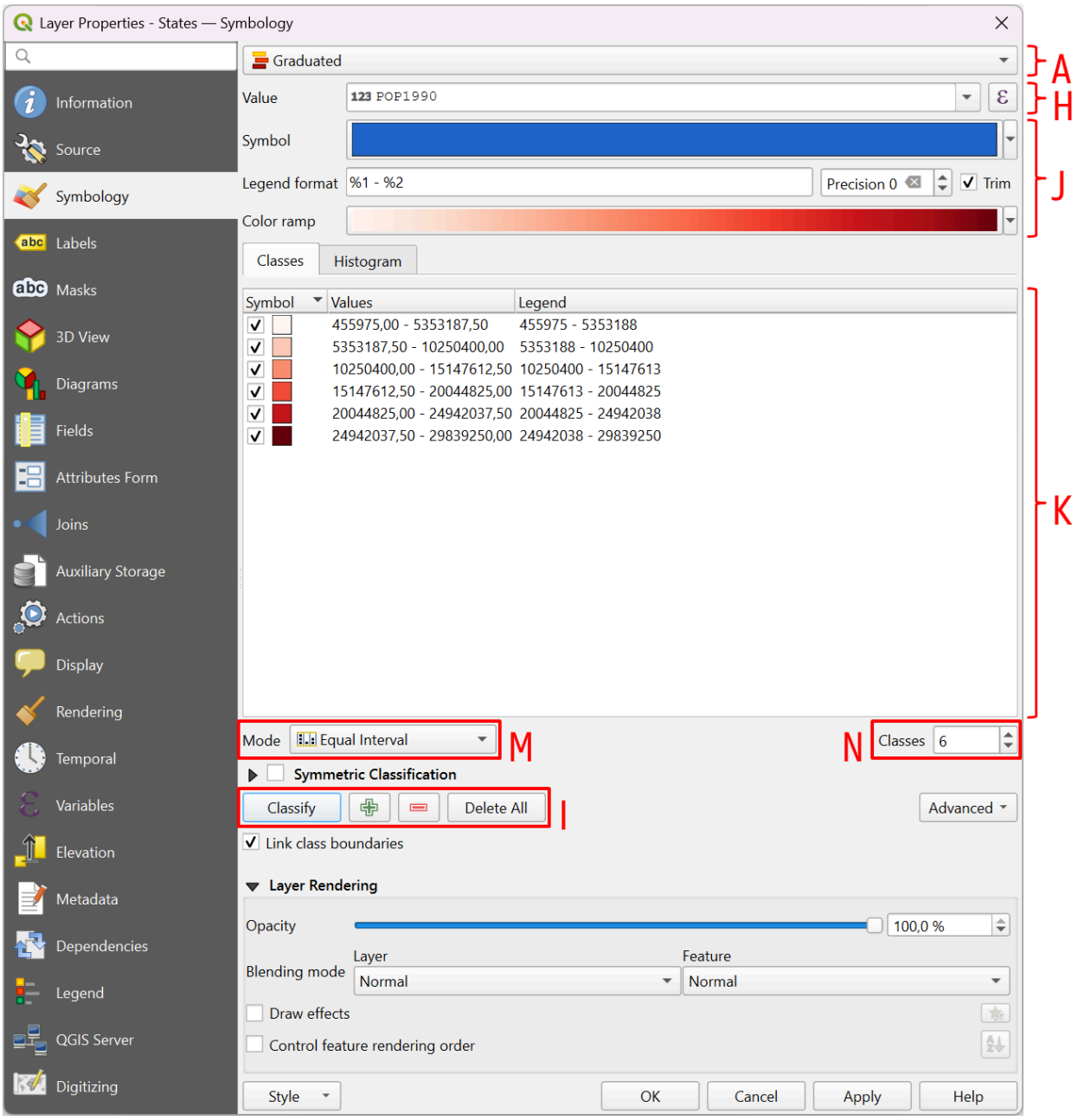
# 4.2.5 Graduated symbology

## 1. Automatic configuration




| | GEOID | STUSPS | NAME | POP1990 | POP2000 | POP2010 |
|---|---|---|---|---|---|---|
| 1 | 06 | CA | California | 29839250 | 33930798 | 37341989 |
| 2 | 36 | NY | New York | 18044505 | 19004973 | 19421055 |
| 3 | 48 | TX | Texas | 17059805 | 20903994 | 25268418 |
| 4 | 12 | FL | Florida | 13003362 | 16028890 | 18900773 |
| 5 | 42 | PA | Pennsylvania | 11924710 | 12300670 | 12734905 |
| 6 | 17 | IL | Illinois | 11466682 | 12439042 | 12864380 |
| 7 | 39 | OH | Ohio | 10887325 | 11374540 | 11568495 |
| 8 | 26 | MI | Michigan | 9328784 | 9955829 | 9911626 |
| 9 | 34 | NJ | New Jersey | 7748634 | 8424354 | 8807501 |
| 10 | 37 | NC | North Carolina | 6657630 | 8067673 | 9565781 |
| 11 | 13 | GA | Georgia | 6508419 | 8206975 | 9727566 |
| 12 | 51 | VA | Virginia | 6216568 | 7100702 | 8037736 |
| 13 | 25 | MA | Massachusetts | 6029051 | 6355568 | 6559644 |
| 14 | 18 | IN | Indiana | 5564228 | 6090782 | 6501582 |
| 15 | 29 | MO | Missouri | 5137804 | 5606260 | 6011478 |
| 16 | 55 | WI | Wisconsin | 4906745 | 5371210 | 5698230 |
| 17 | 47 | TN | Tennessee | 4896641 | 5700037 | 6375431 |
| 18 | 53 | WA | Washington | 4887941 | 5908684 | 6753369 |
| 19 | 24 | MD | Maryland | 4798622 | 5307886 | 5789929 |
| 20 | 27 | MN | Minnesota | 4387029 | 4925670 | 5314879 |

Graduated numeric values are numbers that can be placed on a scale from low to high. Unlike categories such as *city* or *village*, they do not form a limited list of separate values. Attribute fields often contain such **graduated numeric values**, for example the *population* (see POP1990, POP2000 and POP2010 in the example above) or the *area* of a feature. In practice, this means that each feature usually has a different value: it is, for example, highly unlikely that two states have exactly the same number of inhabitants. This kind of data needs a different way of visualising than a limited set of separate categories.




To give a layer a graduated symbology, open its *Layer Properties*, go to *Symbology* and choose *Graduated* in the list at the top (A). For point and line layers, you then choose under *Method* whether the classes differ in **colour** (default) or in **size** (points) / **thickness** (lines); polygon layers do not have this option. The rest of the setup works similarly to a *Categorized* symbology: you choose the attribute field (H) and set a few style options, such as the *Color ramp* (J).

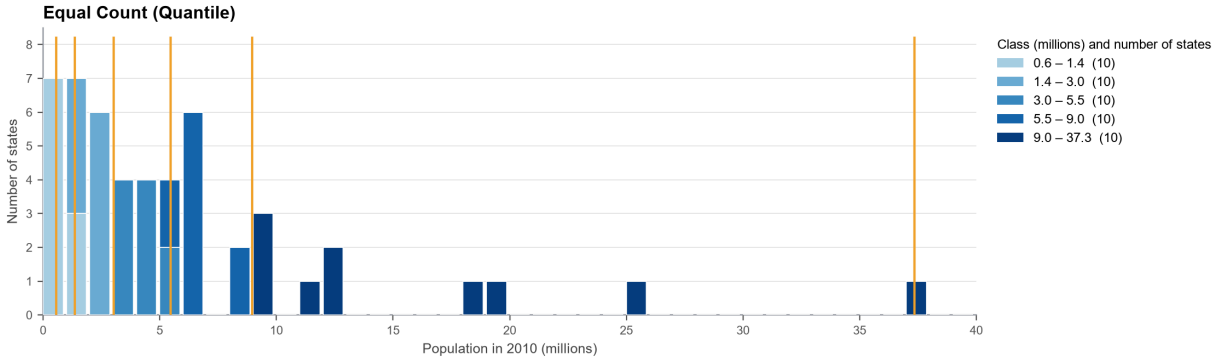
A *Graduated* symbology divides the values into **classes defined by a lower and an upper limit**. All features in the same class get the same symbol. QGIS offers several methods to calculate these classes automatically. You decide how this is done:

- you set **how many classes** you want (N)
- you choose **how QGIS calculates the class limits** under *Mode* (M)

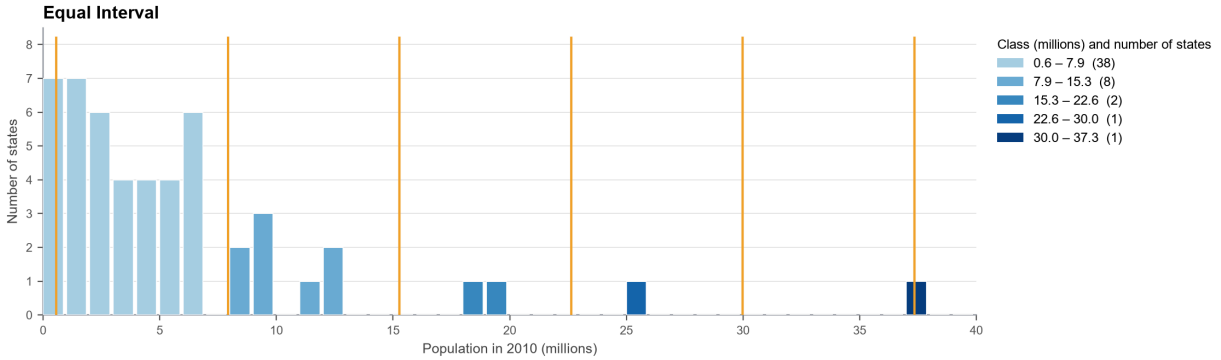
Below, we discuss the options in the *Mode* list.

- **Equal Count (Quantile)**: each class contains (as far as possible) the same number of features. The **advantage** is that no class dominates the map: each symbol appears about equally often. The **downside**

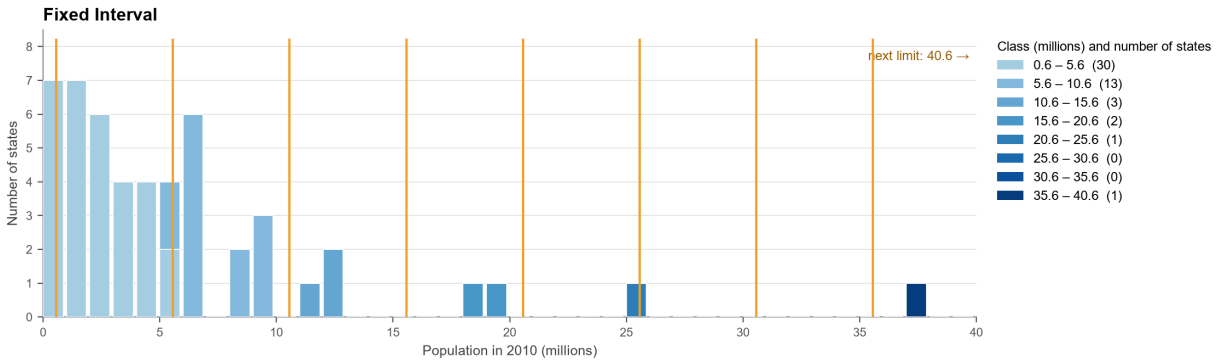
is that the classes can have very different widths: values that are close to each other can end up in different classes, while very different values can share a class. In the example, each class contains 10 states, but the last class runs from 9 to 37.3 million.




- **Equal Interval**: QGIS divides the range of values into classes of the **same width**. It calculates this width by dividing the difference between the highest and the lowest value by the number of classes you asked for. The **advantage** is that each step from one class to the next stands for the same difference in value, which makes the map easy to interpret. The **downside** is that with unevenly spread values, one class can contain almost all features, while other classes stay (nearly) empty. In the example, each class is (37.3 − 0.6 million) / 5 ≈ 7.4 million wide, and 38 of the 50 states end up in the first class.

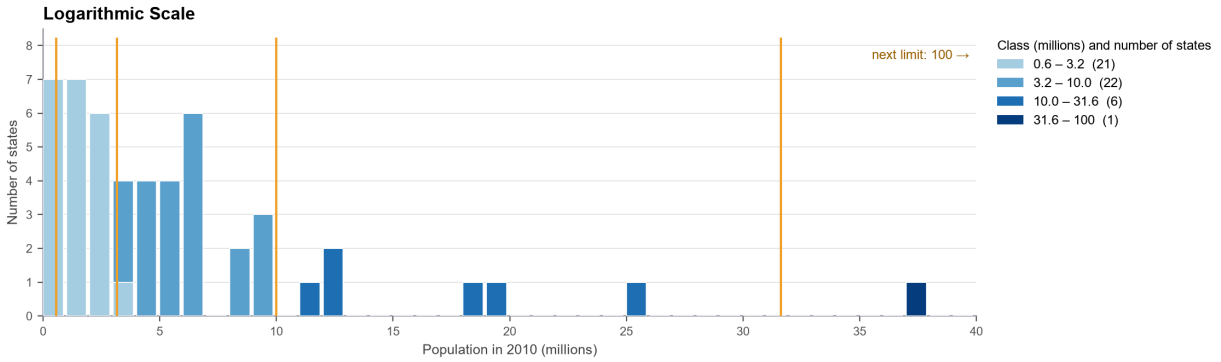



- **Fixed Interval**: this works like *Equal Interval*, but the other way round: you set the width of the classes yourself under *Interval size*, and QGIS calculates how many classes are needed to get from the smallest to the largest value. The number under *Classes* is not used. In the example, an interval size of 5 million inhabitants gives 8 classes, two of which are empty.

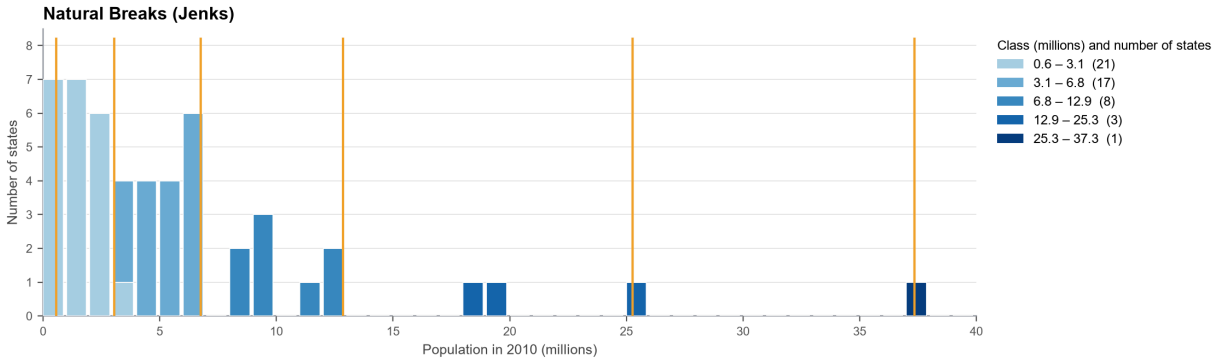



- **Logarithmic Scale**: the class limits follow a logarithmic scale: **each class is wider than the one before**. QGIS places the limits at round values on that scale. As a result, the number of classes can differ from the number you asked for. The **advantage** is that this method suits values that range from very

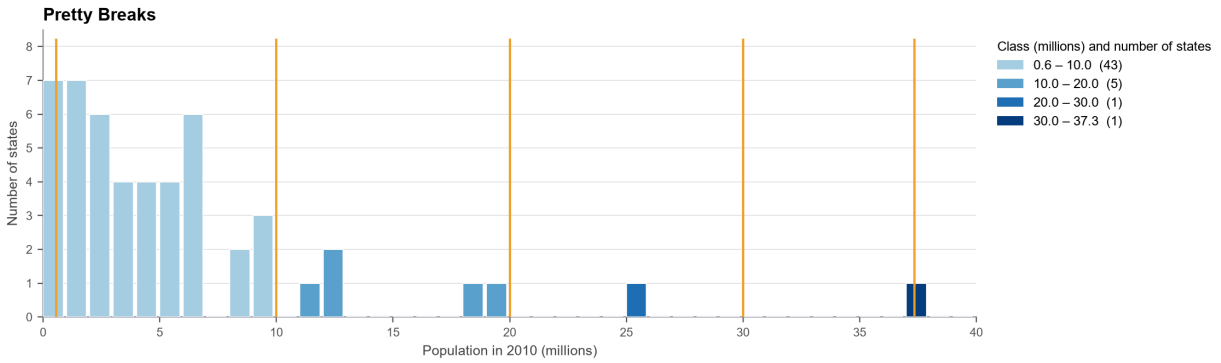
small to very large, with many small values and a few large ones. In the example, QGIS makes 4 classes, with limits at 3.2 million ($10^{6.5}$), 10 million ($10^{7}$) and 31.6 million ($10^{7.5}$); the last class ends at 100 million.




- **Natural Breaks (Jenks)**: QGIS looks for groups (**clusters**) of values that lie close together: it makes the differences between the values *within* each class as small as possible, and the differences *between* the classes as large as possible. The class limits therefore usually fall in the gaps between such clusters. The **advantage** is that the classes follow how the values are actually distributed. The **downside** is that the classes have different widths, which makes the map harder to interpret. In the example, the limits lie at 3.1, 6.8, 12.9 and 25.3 million.

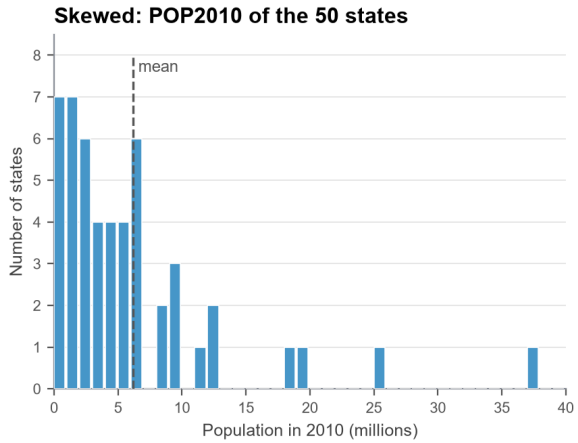


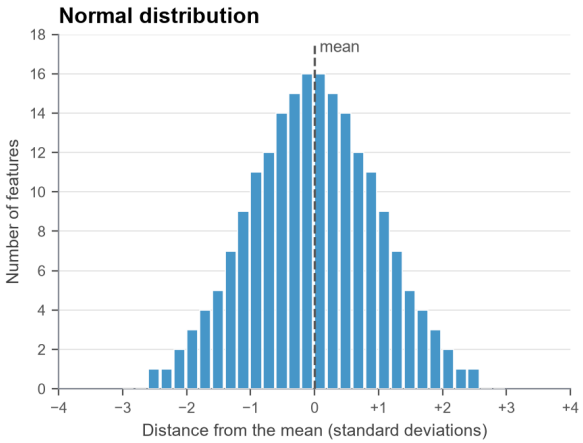

- **Pretty Breaks**: QGIS divides the range of values by the number of classes you asked for, and rounds this class width to an easy-to-read number. The result is a fixed class width with round numbers as limits between the classes. Because of this rounding, the final number of classes can differ from the number you asked for. The **advantage** is a legend that is easy to read. In the example, 36.8 million / 5 ≈ 7.4 million is rounded to 10 million, which gives limits at 10, 20 and 30 million and 4 classes instead of 5.




- **Standard Deviation**: the classes show how far the values lie above or below the mean. The class limits lie at fixed steps from the mean, expressed in standard deviations, for example steps of 1, 0.5 or 0.2 standard deviation. In QGIS, this method only works well when the values are spread roughly

symmetrically around the mean, as in a *normal distribution*. With a strongly *skewed* distribution, such as the populations in the example, this method is in practice unusable.



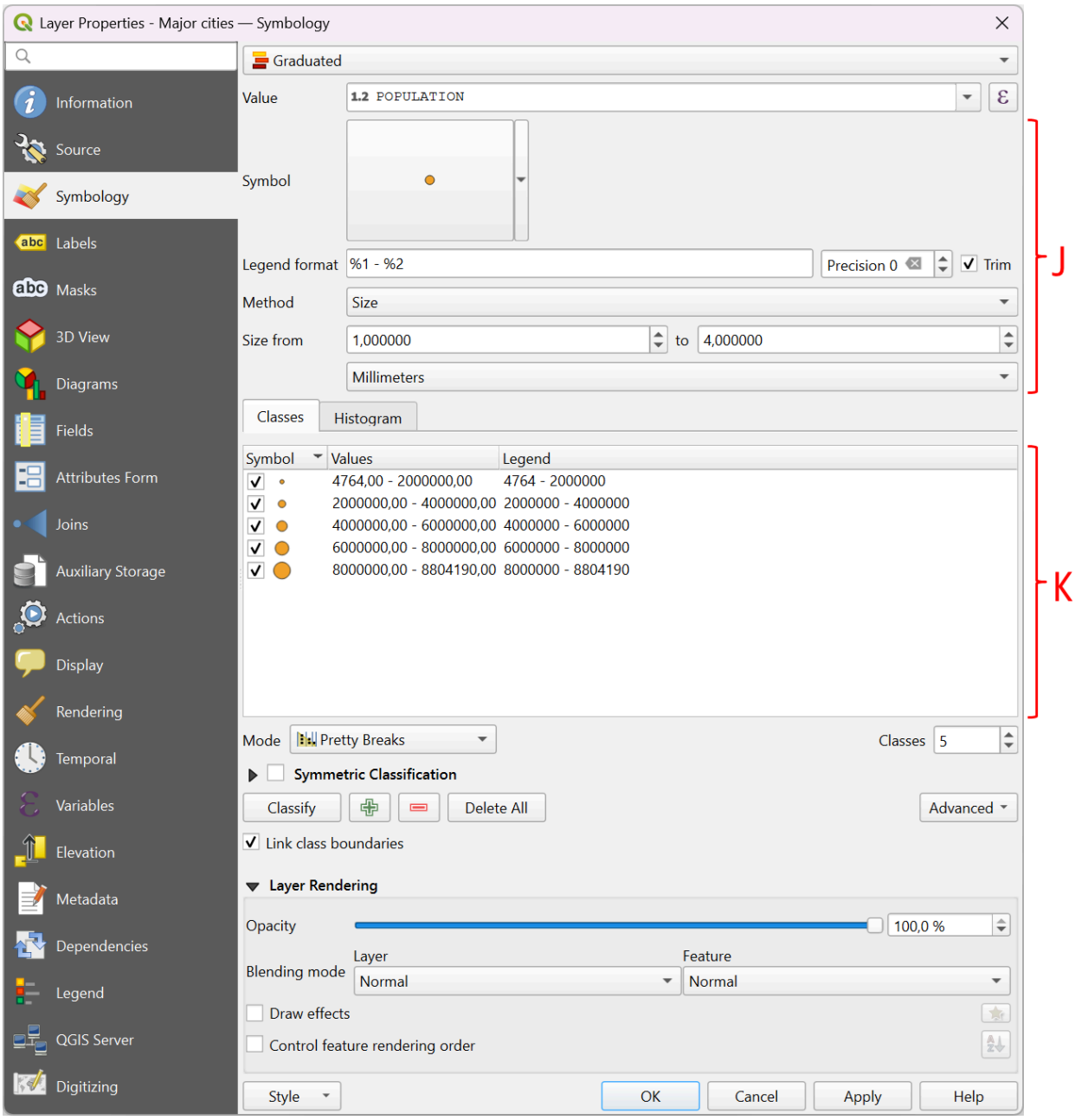




## 2. Manual configuration

You can also **build the classes yourself**. Click the green plus symbol under the list of classes (K): QGIS adds a new class, and you set its lower and upper limit by double-clicking it in the *Values* column. If you started from a calculated classification, you can adjust its limits in the same way. In the *Histogram* tab (next to *Classes*) you can also see how the values are spread, and move the class limits there.

Warning

If you change the number of classes or the *Mode* after adjusting classes by hand, QGIS calculates all classes again, and your manual changes are lost.


Layer Properties - Major cities — Symbology
Information
Source
Symbology
Labels
Masks
3D View
Diagrams
Fields
Attributes Form
Joins
Auxiliary Storage
Actions
Display
Rendering
Temporal
Variables
Elevation
Metadata
Dependencies
Legend
QGIS Server
Digitizing
Graduated
Value
1.2 POPULATION
Symbol
Legend format
%1 - %2
Precision 0
Trim
Method
Size
Size from
1,000000
to
4,000000
Millimeters
Classes
Histogram
Symbol
Values
Legend
4764,00 - 2000000,00
4764 - 2000000
2000000,00 - 4000000,00
2000000 - 4000000
4000000,00 - 6000000,00
4000000 - 6000000
6000000,00 - 8000000,00
6000000 - 8000000
8000000,00 - 8804190,00
8000000 - 8804190
Mode
Pretty Breaks
Classes
5
Symmetric Classification
Classify
Delete All
Advanced
Link class boundaries
Layer Rendering
Opacity
100,0 %
Blending mode
Layer
Feature
Normal
Normal
Draw effects
Control feature rendering order
Style
OK
Cancel
Apply
Help
J
K